RAFAEL HARRY GOMAR DAWSON

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EXPERIENCE

MIT CSAIL – Kellis Lab 08/2025 – Present

Collaborating Researcher

- Led interdisciplinary research on representation learning and embedding-based methods across law, patents, political science, and other applied domains.
- Served as an instructor and lecturer for MIT IAP 6.S189, 6.S187, and MIT Sloan 15.S65, covering applied AI, machine learning, and deployment.

Mantis Present

Founding Team

MAKERS – ITAM 08/2023 – Present

Coordinator, Computer Science Division

- Led and mentored 15 students in 7 development and 9 research projects that integrate computing to benefit the ITAM community, fostering collaboration and technical growth.
- Oversaw recruitment, organization, and participation across the division's 2 areas (Data science and Software Development).

Universo de Leyes Initiative 11/2022 – 2025

Lead Researcher

- Developed an NLP-based information extraction system that generates knowledge graphs to visually represent citational relationships of Mexican legal domains.
- Created a comprehensive annotated dataset to fine-tune a RoBERTa model and train a named entity recognition pipeline using SpaCy. Authored a research paper detailing findings; presented at ICITS 2025.

Spingere 06/2022 – 08/2022, 06/2023 – 07/2023

Research Intern

- Conducted research, planning and prototyping of an NLP solution to automate COSMIC software functional size analysis.

AWARDS & ACHIEVEMENTS

ITAM Hackathon	**ISAC Hackathon**	**INEGI Datatón**
Winner, 2023	Winner, 2025 · Finalist, 2024	Winner, 2025 · Finalist, 2024

ITAM Engineering Academic Excellence Recognition · 2023, 2024, 2025

RELEVANT PROJECTS

Tellus — Geospatial ATM Expansion Planning

Built a geospatial platform that ranks ATM sites for three major banks using local demand, competitor coverage, transaction potential, and learned location representations.

Retail Informality Prioritization Model — INEGI Datatón 2025

Won INEGI Datatón 2025 with an analysis that pinpointed the products, regions, and business segments where policy intervention could have the greatest effect on commercial informality.

- Built a prioritization model from wage, social-security, education, and Economic Census data to diagnose barriers to formalization and match them with targeted policy actions.

SEMOVI — Public Transport Route Digitization

Built a pipeline for Mexico City's mobility authority, SEMOVI, that converts written public-transport route records into digital map data.

- Processed official scanned documents, extracted route descriptions, and reconstructed the corresponding geospatial paths with minimal manual input.

Sustainable Auto Parts Matching

Worked with InvierteSustentable and JetzApp on a catalog-matching system that helps buyers find compatible, lower-impact alternatives to automotive parts.

- Prototyped LLM-based agents that adapt Python preprocessing functions to inconsistent catalog schemas and match equivalent parts across suppliers.

PUBLICATIONS & SEMINARS

Comparative Analysis of Information Extraction Techniques in the Mexican Legal System

F. Valdés, R. H. Gomar and D. A. Méndez

- Published in Proceedings by Springer in Lecture Notes in Networks and Systems 2025 and presented at the 8th International Conference on Information Technology & Systems, held at Politécnico Nacional

Universo de Leyes: Aplicando el Procesamiento del Lenguaje Natural para vincular textos jurídicos con un enfoque integral

- Seminario de Ingeniería de Software y Sus Implicaciones, Nov 2024, Facultad de Ciencias, UNAM
- Seminario Recursivo, Mar 2024, División Académica de Ciencias de la Computación, ITAM

Y si pudiéramos entender el contexto a través de los datos? | Player2Vec

- Hudl Statsbomb, ITAM Sports Analytics Conference 2025